

SHREEHARI AVADHANI

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Summary

CSE student with foundations in Python, C++, SQL and data structures and algorithms. Experienced in building end-to-end AI/ML and backend systems using Python, FastAPI and PostgreSQL, with hands-on experience in LLMs, RAG, embeddings and AI agents. Interested in building and evaluating reliable AI systems through systematic analysis and debugging.

Technical Skills

Languages: Python, C++, SQL

Data & ML: Pandas, NumPy, Scikit-learn, Supervised & Unsupervised Learning, AI Evaluation

AI/GenAI: LLMs, Generative AI, Agentic AI, RAG, Embeddings, Prompt Engineering, Sentence Transformers, LangGraph, RAGAS, Chroma

Backend & Tools: FastAPI, REST APIs, Docker, Docker Compose, Git, GitHub, LangSmith, PyMuPDF, Groq

Databases: PostgreSQL, pgvector, Redis, Supabase

Core CS: Data Structures & Algorithms, OOP, DBMS, Operating Systems, Computer Networks

Education

Nitte Meenakshi Institute of Technology

B.E. Computer Science and Engineering

Oct 2023 – July 2027

Bengaluru, KA

- CGPA: 8.05

Projects

Autonomous arXiv Paper Digest & QA Agent | 🤖 | *Python, LangGraph, Chroma, Sentence Transformers, Groq* **2026**

- Built a stateful research agent using Python and LangGraph that accepts arXiv IDs, URLs, or research topics and autonomously retrieves and processes research papers.
- Implemented PDF parsing, chunking, MPNet embeddings, and persistent Chroma storage, with targeted retrieval for structured paper briefings covering problems, methods, results, and limitations.
- Implemented hybrid semantic and lexical retrieval for grounded multi-turn QA, with retrieval gating to prevent LLM calls when relevant paper evidence is not found.
- Added explicit LangGraph state management, QA looping, paper switching via `/paper`, and failure handling for invalid papers, parsing errors, large PDFs, and embedding failures.

Multitenant RAG System with Semantic Caching | 🤖 | *Python, FastAPI, PostgreSQL, pgvector, Redis, Docker* **2026**

- Built and evaluated a production-oriented multitenant AI retrieval system using Python, PostgreSQL and pgvector, with tenant isolation enforced at the SQL layer.
- Implemented hybrid retrieval using BM25 and pgvector dense search with Reciprocal Rank Fusion and cross-encoder reranking to improve document relevance.
- Evaluated retrieval and response quality using RAGAS, achieving **0.75 Faithfulness**, **0.75 Answer Relevancy**, **0.83 Context Precision**, and **0.73 Context Recall**.
- Containerized the system using Docker Compose and implemented Redis semantic caching, IVFFlat vector indexing, health checks, and auto-recovery.

NMIT-GPT – Institutional AI Chatbot | 🤖 | *Python, FastAPI, PostgreSQL, pgvector, RAG*

2026

- Built an end-to-end AI assistant using Python and FastAPI for institutional academic queries, integrating PostgreSQL, pgvector, Sentence Transformers, and Groq LLMs.
- Implemented semantic vector retrieval, SQL-based structured querying, and LLM-based intent detection to route structured and unstructured requests.
- Implemented multi-turn session handling and pronoun resolution to maintain context across follow-up queries.
- Integrated structured and unstructured knowledge retrieval to support academic queries across institutional data.

Leadership

Esports Co-Lead, Oasis (NMIT Events) | Coordinated institute-level esports tournaments, managing registrations, scheduling, brackets, match execution, and participant communication.